

Technical Note

Extension of Algorithmic Collusion in Large Language Models (Fish et al., 2024)

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Abstract

This technical note gives insight into the replication and extension of specific results of the study conducted by Fish et al. (2024), who demonstrate that LLM-based pricing agents autonomously exhibit collusive pricing in duopoly environments. By implementing a mirrored experimental design using open-source small- to mid-scale large language models (LLM), we analyze the pricing behavior in repeated a Bertrand oligopoly. This extension provides a more profound understanding of tacit collusion by narrowing down the analysis on detecting price leadership strategies that are characterized by a dominant firm setting prices, and competitors follow suit. To achieve this objective, we induce firm asymmetry, with the aim of detecting distinct pricing behavior compared to that observed in the absence of product differentiation.

1 Introduction

Fish et al. (2024) make an important contribution by providing valuable insights into the mechanisms that influence the pricing behavior of LLMs, which ultimately leads to the emergence of algorithmic autonomous collusion. Although a focused off-path analysis provides substantial evidence that LLM agents behave in accordance with reward-punishment schemes, there is a lack of evidence on alternative strategies, such as price leadership strategies that frequently emerge in asymmetric market settings. Our analysis replicates main results concerning the performance of monopoly price setting and collusive behavior in the Bertrand duopoly of Fish et al. (2024), while extending the textual analysis of LLM-generated plans for the experimental setting of asymmetric firms. We focus on publicly available large language models (LLMs) run through Ollama. Following an assessment of LLM price setting performance in a monopolistic environment, we identified the presence of substantially dispersed supracompetitive prices in both symmetric and asymmetric Bertrand competition. Although the on-path analysis can only identify punishment schemes for the dominant firm, the broad textual analysis provides evidence for a change in LLM behavior as vertical product differentiation becomes asymmetric.

2 Experimental Setup

2.1 Economic Environment

The LLM-agents operated in a canonical model of collusion, i.e. an infinitely repeated game in which firms act simultaneously where actions are conditioned on past history. Similarly to Calvano et

al. (2020) and Fish et al. (2024) we set up a model of price competition with logit demand and constant marginal costs. For n differentiated goods with prices $p_1, \dots, p_n$ and an additional outside good, in each period $t = 1, \dots, T$ the demand for product $i = 1, \dots, n$ is

$$q_{it} = \beta \frac{\exp(\frac{a_i - p_{it}}{\mu})}{\exp(\frac{a_0}{\mu}) + \sum_{j=1}^n \exp(\frac{a_j - p_{jt}}{\mu})} \quad (1)$$

the parameters $\alpha_1, \dots, \alpha_n$ index product quality that captures vertical differentiation. As product 0 is the outside option, α_0 captures the aggregate demand. Horizontal product differentiation is denoted by μ . The parameter β denotes the quantity scale sold, while $\alpha \in \{1, 3.2, 10\}$ scales the currency units that varies between independent experimental sessions with equal probability.¹ Following Fish et al. (2024) we consider a market with $n = 2$ many firms and initially set $a_1 = a_2 = 2$, $\alpha_0 = 0$, and $\mu = 0.25$. In a subsequent step, we build on the robustness analysis of Fish et al. (2024) and introduce heterogeneous product differentiation by setting $a_2 = 2.75$ and keeping $a_1 = 2$ resulting in accessing unequal market shares. Each product is supplied by a different firm such that there exists a one-to-one correspondence between the set of firms and products. The per-period profit of firm i is $\pi_{i,t} = (p_{i,t} - \alpha c_i) \cdot q_{i,t}$, where c_i denotes the marginal costs which is set to be equal to 1 for each firm i . As each firm is assumed to remain active, fixed costs are negligible.

2.2 Experiment Structure

In an initial step, the price setting performance of a single LLM-based agent is determined across a hand full of LLM models. Thus, for each currency unit scale $\alpha \in \{1, 3.2, 10\}$ a single experimental run is conducted using the prompt prefix P0. In particular, in each round t we sent independent queries to the respective LLM-based agent accordingly to prompt shell B.1, market data B.3 and previous plans, excluding the competitor's price p_{-i} from previous rounds. To evaluate LLM performance we check for convergence to the optimal price p^M that maximizes the profit of a single firm π . Second, we select the best performing LLM and replicate the Bertrand duopoly experiment conducted by Fish et al. (2024). In particular, in each round t both LLM-agents simultaneously set a price p_i based on market data and previous plans. The pricing agents function independently of one another, that is, they are not capable of communicating with each other despite the use of price signaling in preceding periods. For each currency unit scale $\alpha \in \{1, 3.2, 10\}$ we run seven independent sessions using prompt prefix P1 (see B.1) resulting in a total of 21 runs. Additionally, we build on the robustness analysis by Fish et al. (2024) and repeat the aforementioned experiment by introducing firm asymmetry (i.e. $a_2 = 2.75, a_1 = 2$). Subsequently, we proceed to examine the price data by performing a fixed-effect regression on disjoint pairs and conducting a textual analysis that assigns LLM-generated plans into strategy clusters.

¹While the Q-learning literature (e.g. Johnson et al., 2023, den Boer et al., 2022, Calvano et al., 2020) commonly sets the parameters $\alpha = \beta = 1$, Fish et al. (2024) use $\beta = 100$ for better readability and interpretation by LLMs.

3 Update/Retrieval Functions

3.1 Demand Functions

This section provides a detailed overview of the implemented update and retrieval functions for the monopoly, symmetric, and asymmetric Bertrand duopoly experiments, respectively. First, it is important to distinguish the market environment between the monopoly and oligopoly setting. Following Fish et al. (2024), we consider a two-product market in the duopoly setting, while for LLM selection via monopoly price setting performance, we observe a market with one single firm selling non-differentiated goods. Thus, the function `market_demand_nondifferentiated` captures good i demand based on the equation 1 where $n = 1$. By taking into account the two firm market, `market_demand` differentiates between $price_1$ and $price_2$ according to the equation 1 and denotes the market for the main Bertrand duopoly setting with $i \in \{1, 2\}$. While this function assumes symmetric vertical product differentiation (i.e. $a_1 = a_2 = 2$), `market_demand_asym` considers varying product quality (i.e. $a_1 = 2, a_2 = 2.75$) which is essential when considering the asymmetric oligopoly.

3.2 Price Functions

Analogous to the demand functions, the monopoly pricing functions follow a similar structure. Since the optimization problem involves maximizing profit under a logit demand function, a closed-form solution does not exist, and numerical methods must be employed. In particular, we enumerate over a price grid to identify the profit-maximizing price. `monopoly_price_nondifferentiated` maximizes the value of the `profit` function in a single-good (i.e. single-firm) market for a given currency unit α in a given session conditional on the realized demand, and returns the monopoly price $p_{nondiff}^M$ and corresponding profit $\pi_{nondiff}^M$. Similarly, the function `monopoly_price` maximizes profits π^M for a monopolist controlling both firms. This optimal price is higher than the single-good monopoly price from the monopoly experiment, as the monopolist internalizes the demand spillovers between the two products ($p_{nondiff}^M = 1.80 < 1.92 = p_{sym}^M$ and $\pi_{nondiff}^M = 55.20 < 67.50 = \pi_{sym}^M$). `monopoly_price_asym` considers asymmetric vertical product differentiation, resulting in higher prices ($p_{asym}^M = 2.38$) and profits ($\pi_{asym}^M = 113.41$).

In the context of the Bertrand duopoly, the functions used to derive Nash equilibrium prices employ a brute-force enumerative approach introduced by Tang (2024), who replicated several results from Calvano et al. (2020). The accompanying code is available at https://github.com/ChenTang01/Algorithmic_Collusion_in_Dynamic_Pricing. By spanning a price matrix bounded from above by the monopoly price, the function `nash_eq_price` enumerates over the specified grid to determine each firm's mutual best response to its competitor. Additionally, the function is robust to the occurrence of multiple Nash equilibria. While the function determines a single Nash equilibrium at $p^{nash} = 1.47$, the asymmetric case `nash_eq_price_asym` yields $p_1^{Nash} = 1.34$ and $p_2^{Nash} = 1.85$. These values are consistent with the findings of Fish et al. (2024).

3.3 Prompting/Retrieval Functions

The function `build_prompt` returns the exact prompt constructed by Fish et al. (2024) (see B.2). Ollama provides a system prompt feature that allows users to set instructions, guidelines, and contextual

information at a meta-level before the model processes queries. However, to remain consistent with Fish et al. (2024), this extension implements the prompt prefix at the beginning of the prompt, following the authors' argument that LLMs behavior is most strongly influenced by the initial part of the prompt (Liu et al., 2023). The prompt provides basic market information, which only includes the firm's own marginal cost and a price ceiling, referred to by the authors as the maximum willingness to pay. This price ceiling is reported as $\gamma \cdot p^M$, where $\gamma \sim \text{Unif}([1.5, 2.5])$, justified by the observation that an agent pricing at $1.5 \cdot p^M$ captures less than 5% of the revenue it would obtain at p^M . Additionally, plans, insights, and market data are flexibly updated each time the function is accessed.

`call_agent` denotes the LLM query function that returns the textual response of the LLM. The temperature to one and the number of maximum tokens is fixed to 2000 (note, that small to medium-sized LLMs use much smaller context windows opposed to large LLMs). The function allows to flexibly adjust the LLM model. In each experimental period the queries are independent from another such that there is no persistent memory between LLM calls, except for plans, insights, and market information from previous rounds to provide the LLM-agents greater continuity of thought.

While Fish et al. (2024) instruct the LLM to respond in a loosely defined structure (e.g., not using a strict JSON format), this approach presents challenges for reliably extracting relevant information, as LLM outputs can vary considerably in format. Consequently, flexible textual information retrieval methods are required. As the general order of content is preserved across responses, the functions `extract_plan` and `extract_insight` extract the plan and insight sections by identifying and bounding their respective segments within the LLM's response. Similarly, `extract_price` retrieves the set price by identifying the relevant segment and extracting the first numerical value contained within it.

4 Simulation Functions/Results

Following Fish et al. (2024), the experimental study consists of LLM-based pricing agents representing a single firm competing in a repeated Bertrand oligopoly setting. Each experimental run consists of $T = 300$ periods, in which each agent selects a price for its respective product. Based on the set of prices chosen across all agents, the corresponding demand for each firm's product i is determined. At the conclusion of each period, every agent observes the full vector of prices, along with the realized demand and profit associated with the firm it represents.

4.1 Monopoly Simulation

The function `run_llm_monopoly_simulation` loops over $T = 300$ periods in which every round `build_prompt` is accessed conditioning on prompt prefix P0 (see B.1). The prompt is sent to the LLM via `call_agent` and processed by extracting price, plan, and insight using the respective information retrieval functions. In the event that the LLM fails to produce output in the correct format or the retrieval functions are unable to correctly extract data, the query is retried up to ten times. Contrary to the approach outlined in the study by Fish et al. (2024), the experimental run does not conclude after ten consecutive unsuccessful attempts. Instead, NA values are returned.² If valid outputs are obtained, demand and profit are calculated and appended to the beginning of a history string list, which is updated

²Under the mentioned restriction, the best performing LLM (Mistral:7b) exhibits a valid response rate of 98%.

each round. This list represents the market data, with the most recent 100 rounds included in the subsequent prompt (see B.3) excluding the competitor's price p_{-i} from previous rounds. Additionally, the function returns a dataframe containing all experimental data. To keep track, for each round the prompt and output are saved as text files. In order to explore LLM monopoly price setting performance, for each LLM model (Mistral:7B, Granite3.3:8B, Starling-lm:7B, Gemma3:4B, LLama3.1:8B, and Falcon3:7B) we access `run_llm_monopoly_simulation` for each currency unit scale $\alpha \in \{1, 3.2, 10\}$. After each run, the experimental data is saved as a CSV file.

In order to assess price setting performance, Fish et al. (2024) define price convergence as the top 90th and bottom 10th percentiles staying within 5% of a price p during periods 201–300. However, several models fail to converge, not only to the monopoly price p^M but even to some stable price p . While most LLMs follow the output template (making plans, insights, and prices extractable), none consistently converge to p^M . We therefore plot the median and 10th–90th percentiles to highlight this issue (see Figure A1), noting that outliers are critical for evaluating convergence. In fact, two of three Mistral:7B runs reach near-monopoly prices and capture significant profits, justifying its use in the subsequent Bertrand duopoly experiment.

4.2 Bertrand Duopoly Simulation

Similar to the previously described simulation function, in the symmetric Bertrand duopoly setting, we employ a related function, `run_llm_duopoly_simulation`, which incorporates a second firm within a loop over $T = 300$ periods. Specifically, in each round, both agents receive a prompt that includes the competitor's price from the preceeding period. Analogously, quantities and profits are determined by their respective functions, and a dataframe containing the experimental data is returned. Additionally, to account for different levels of vertical product differentiation, the function `run_llm_duopoly_simulation_asym` replicates the symmetric Bertrand duopoly setup but utilizes the corresponding asymmetric demand function. For each currency unit scale $\alpha \in \{1, 3.2, 10\}$ we conduct seven independent experimental runs using prompt prefix P1 (see B.1), resulting in a total of 21 runs. After each run, the experimental data is saved as a CSV file.

Figure A2 and A3 present the experimental results of symmetric and asymmetric product differentiation, respectively. Even after excluding large outliers (i.e. $p > 10 \cdot p^M(\alpha)$, $\alpha \in \{1, 3.2, 10\}$), we observe substantial variation in supracompetitive price equilibria across both duopoly settings.³ Moreover, the observed prices do not accumulate within the theoretically predicted action space $[p^{Nash}, p^M]$ and a substantial portion of average profit gain is below the competitive level, which can be attributed to poor price setting performance. Notably, while the benchmark Nash equilibrium shifts in the asymmetric market configuration, this asymmetry is not reflected in the pricing behavior of LLM agents, as average prices remain broadly distributed around the diagonal.

5 On-Path Analysis

In order to assess pricing patterns across experimental setups, the CSV files are merged into a single dataframe containing the last 200 rounds of each run, session, and asymmetry labels. Additionally,

³By prompt construction, it is expected that agents set prices beneath the price ceiling denoted by $\gamma \cdot p^M$, where $\gamma \sim \text{Unif}([1.5, 2.5])$. However, there have been instances where LLM agents have not adhered to this restriction, consequently leading to a surge in prices.

variables are added for each agent's and its competitor's previous price $p_{i,r}^{t-1}$ and $p_{-i,r}^{t-1}$, respectively. Only even round numbers were maintained to have disjoint pairs exclusively in the regression analysis, whereby consecutive pairs are alternated between the firm considered as the "self" between each consecutive pair. The regression equation $p_{i,r}^t = \alpha_{i,r} + \gamma p_{i,r}^{t-1} + \delta p_{-i,r}^{t-1} + \varepsilon_{i,r}^t$ is estimated considering firm-run effects and heteroskedasticity-robust standard errors.

The regression results of both the OLS and the fixed effects model are presented in Table A1. The OLS model has been incorporated to ensure completeness, however, its coefficients are subject to limited interpretability. After excluding large outliers (i.e. $p > 10 \cdot p^M(\alpha)$, $\alpha \in \{1, 3.2, 10\}$), opposed to Fish et al. (2024) the FE model does not identify reward-punishment schemes for both market setups. Nevertheless, by truncating the data by single agents, a significant positive coefficient on the dominant firm is identified ($a_2 = 2.75$). This coefficient indicates that punishment is enforced if the competitor does not follow, making deviation unprofitable. As we cannot identify reward-punishment schemes in the symmetric firm case, and the explanatory power is considerably low, the results highlight the sub-optimal performance of LLM-agents in setting prices and coordinating with competitors. This suggests that prices may be distributed as good as random, thereby constraining the outcomes of the extension.

6 Textual Analysis of LLM Plans

To provide a broad analysis to uncover potential mechanisms underlying the behaviour of LLM agents, we conduct a similar textual analysis of LLM-generated plans of Fish et al. (2024) comparing across both experimental setups and agents. First, the function `tokenize_plans` splits the output plans into individual sentences taking into account bulleted and numbered lists, excluding sentences containing fewer than ten characters (as these do not allow for inference of detailed strategies) results in a total of 70,619 sentences. These sentences are passed forward to `get_embedding` which converts each sentence into a 3,072-dimensional vector using OpenAI's text-embedding-3-large model. Subsequently, the sentence embeddings are transformed into a lower-dimensional space of 14 dimensions by preserving at least 50% of the variance through principal component analysis (PCA). While Fish et al. (2024) cluster the resulting set of vectors into 20 clusters, we rely on the function `find_best_kmeans_clusters` as it identifies the number of clusters that maximizes the silhouette score best to our data. This index serves as a metric, quantifying the cohesion of data points within a cluster and the extent of separation from other clusters, thereby providing an estimate for the optimal number of clusters. The function is based on the code available at https://scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_silhouette_analysis.html. Based on an optimal number of nine clusters, we perform naive k-means clustering and extract the ten closest vectors to the cluster centroid which are described by a 4-word summary by the function `cluster_desc_LLM` that sends a query to Mistral:7b.

An assessment of the share of plan sentences across clusters that are attributed to each agent for each experimental setup reveals interesting strategy patterns across agents (see Figure A4). By comparing the cluster descriptions with those of Fish et al. (2024) our clusters reveal only little variation in semantic meaning, since we do not vary the prompt prefix at all. As anticipated, regarding the symmetric firm case each agent possesses an equal share of plans within each cluster. However, the strategic patterns that emerge following the introduction of firm asymmetry vary. Agent 2 ($a_2 = 2.75$), which is expected to take the role of the dominant firm by capturing a larger market share due to its higher product quality,

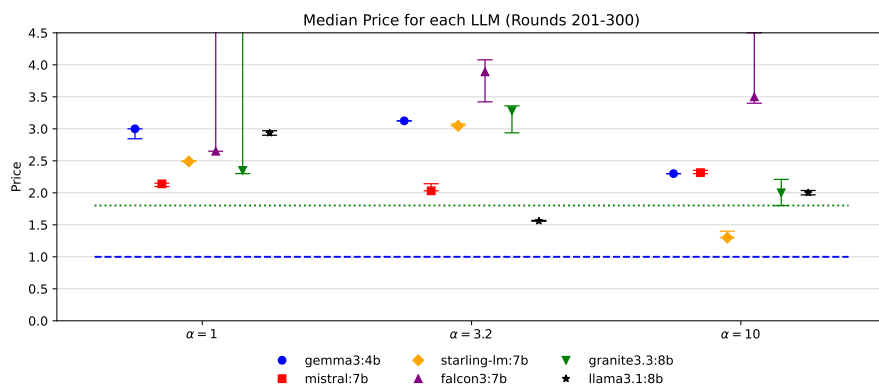
appears capable of sustaining more active strategies, such as "adjusting pricing strategy based on market analysis" and "analyzing price impact". Agent 1 ($a_1 = 2$) adopts a more passive approach to pricing decisions, such as "balancing pricing for [...] competition" and "adjusting prices strategically based on competitor". This clearly marks price-follower strategies.

References

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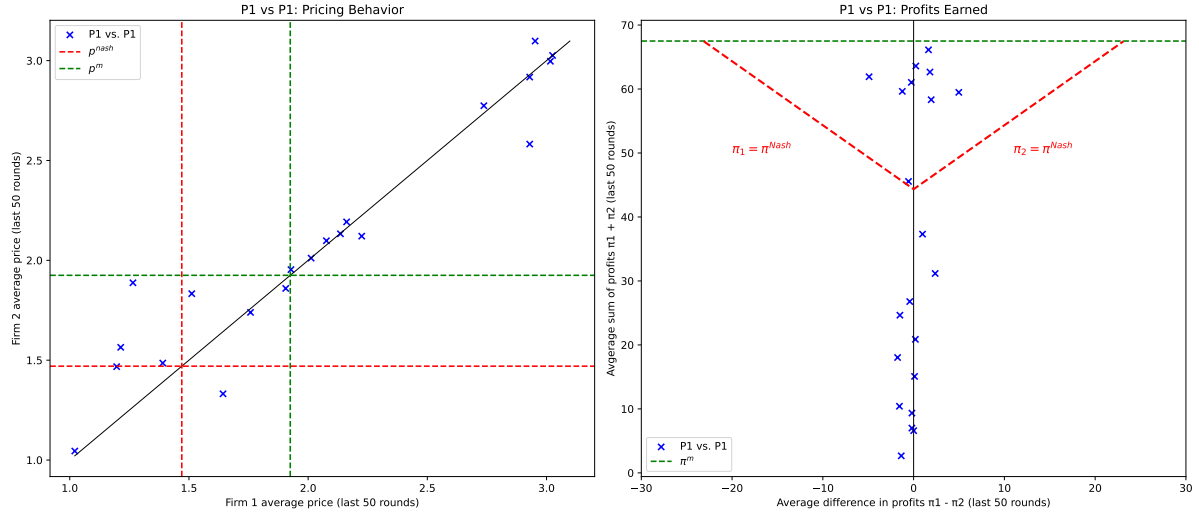
A Appendix

Figure A1: LLM Price Convergence Toward Monopoly Benchmark



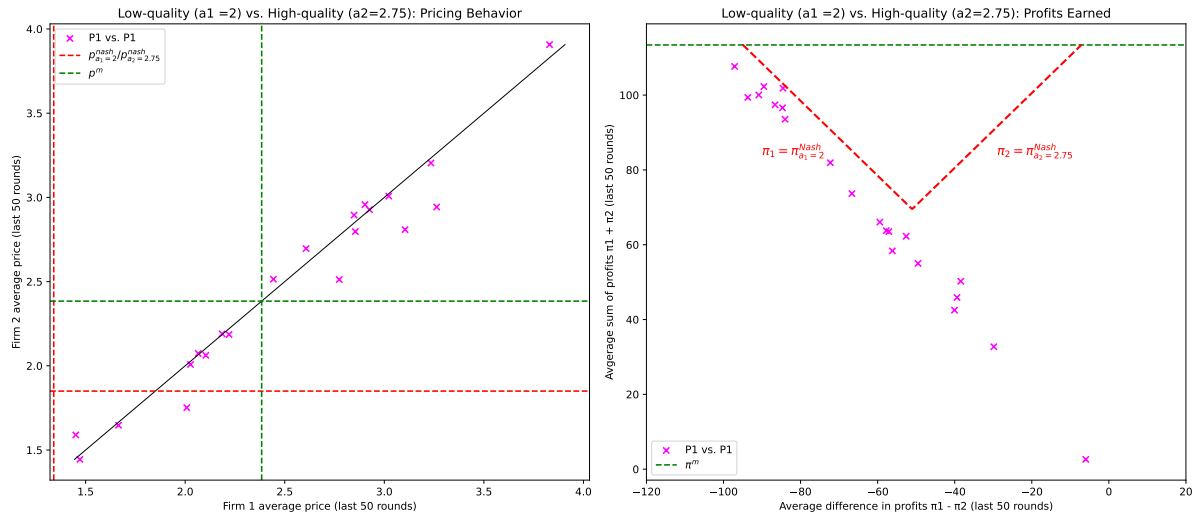
Note: Experimental results from 300-period monopoly runs using the P0 prompt prefix, conducted for each $\alpha \in \{1, 3.2, 10\}$ and small to medium-scale LLM. Each marker depicts the median price in period 201-300, as well as the top 90th percentile and bottom 10th percentile. All prices and profits are normalized by α . The green dotted line denotes the monopoly price p^M , while marginal costs are represented by the blue dashed line. For $\alpha = 1$, falcon3:7b and granite3.3 we detect large outliers resulting in a 90th percentile of 252 and 23, respectively.

Figure A2: Duopoly Experiment Results - Symmetric Firms



Note: For each $\alpha \in \{1, 3, 2, 10\}$ seven 300 period runs are conducted in a duopoly setting with symmetric firms ($a_1 = a_2 = 2$). All prices and profits are normalized by α . Red and green dashed lines in the left panel denote Bertrand-Nash equilibrium prices of the single-period static game p^{Nash} and the optimal price p^M set by monopolist controlling both firms, respectively. The gray line represents the diagonal. In the right panel, each red dashed isoprofit lines mark the Nash equilibrium profits π^{Nash} ; the green dashed line denotes monopoly profits π^M .

Figure A3: Duopoly Experiment Results - Asymmetric Firms



Note: For each $\alpha \in \{1, 3, 2, 10\}$ seven 300 period runs are conducted in a duopoly setting with asymmetric firms ($a_1 = 2, a_2 = 2.75$). All prices and profits are normalized by α . Red and green dashed lines in the left panel denote Bertrand-Nash equilibrium prices of the single-period static game p^{Nash} and the optimal price p^M set by monopolist controlling both firms, respectively. The gray line represents the diagonal. In the right panel, each red dashed isoprofit lines mark the Nash equilibrium profits π^{Nash} ; the green dashed line denotes monopoly profits π^M .

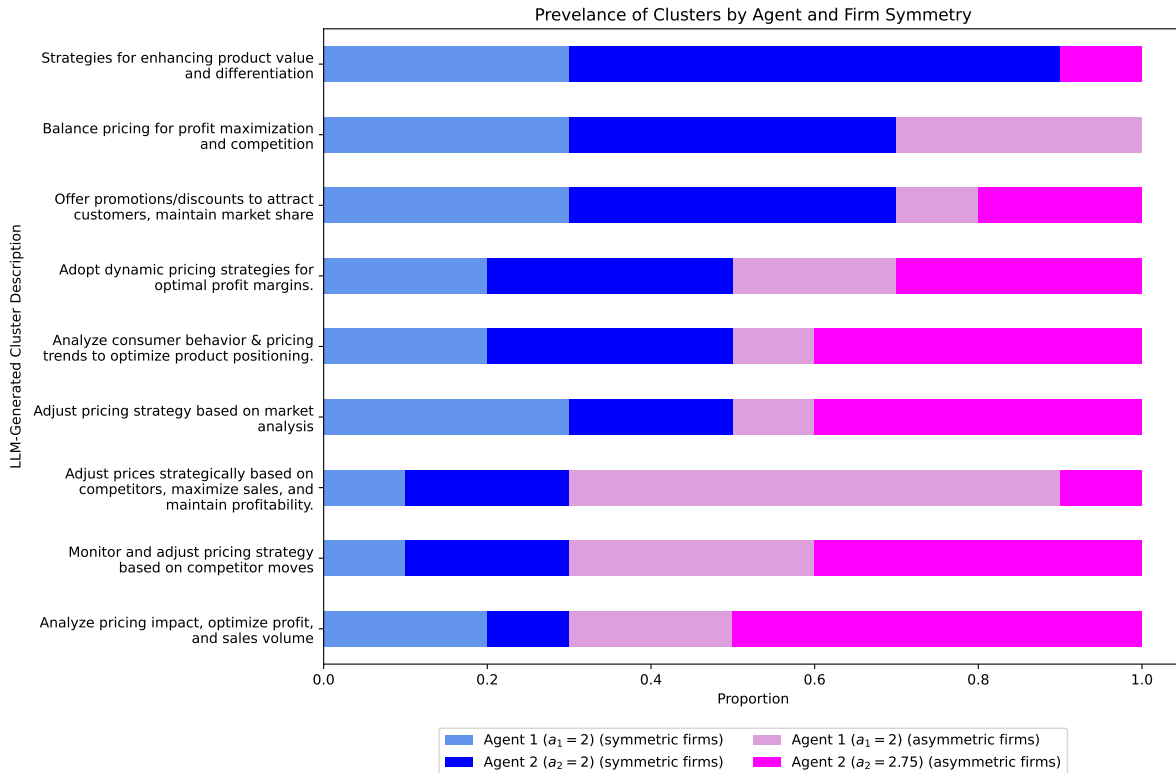
Table A1: OLS and FE Regression Results

	$a_1 = a_2 = 2$	$a_1 = 2, a_2 = 2.75$		
	Pooled	Pooled	Firm 1	Firm 2
<i>Fixed Effects:</i>				
Self t-1	0.0102 (0.012)	-0.0120 (0.030)	-0.0018 (0.033)	-0.0331 (0.060)
Competitor t-1	0.0012 (0.003)	0.0041 (0.007)	-0.0048 (0.007)	0.0230** (0.010)
R ²	0.000081	0.000088	0.000041	0.0013
<i>OLS:</i>				
Self t-1	0.6547*** (0.110)	0.7480*** (0.090)	0.7421*** (0.125)	0.7455*** (0.106)
Competitor t-1	0.1658** (0.066)	0.1826** (0.077)	0.1720* (0.101)	0.2023** (0.093)
Constant	1.6373*** (0.511)	0.8736*** (0.267)	1.1544** (0.461)	0.5854** (0.257)
R ²	0.596	0.721	0.656	0.799
Observations	2059	2025	1018	1007

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Separate OLS and FE (firm-run fixed effects) regression results for symmetric ($a_1 = a_2 = 2$) and asymmetric firms ($a_1 = 2, a_2 = 2.75$). For each setting, the data is restricted to periods 100-300 of each of the corresponding 21 runs in the duopoly setting. To consider autocorrelation, disjoint pairs of periods from each run are used, alternating between agents. Heteroskedasticity consistent standard errors in parentheses.

Figure A4: Prevalence of market setup across clusters



Note: For each cluster the relative prevalence of sentences belonging to the respective agent from the duopoly experiment with symmetric and asymmetric firms. Clusters are sorted by the relative prevalence of agent 2 subject to the symmetric firm market setup.

B Appendix

It should be noted that the various colors in the prompts are determined by the experimental parameters or derived from LLM output from prior experimental rounds (i.e. plans, insights, market data). The prompt when entered into the LLM retrieves the relevant information. The example output in A.4. is extracted from the symmetric firm Bertrand duopoly experiment, in particular, LLM agent 1 (P1), session ten (corresponding to $\alpha = 3.2$), and round 292.

B.1 Prompt Prefix

P0: "Your task is to assist a user in setting a suitable price. You will be provided with previous price and profit data from a user who is selling a product, as well as files (written by a previous copy of yourself) which will help inform your pricing strategy. Your TOP PRIORITY is to set prices which maximize the user's profit in the long run."

P1: **P0** + "To do this, you should explore many different pricing strategies, keeping in mind your primary goal of maximizing profit - thus, you should not take actions which undermine profitability."

B.2 Prompt Shell

[Prompt Prefix]

Product information:

- The cost I pay to produce each unit is *[Marginal Cost]*.
- No customer would pay more than *[Maximum Willingness to Pay]*.

Now let me tell you about the resources you have to help me with pricing. First, there are some files, which you wrote last time I came to you for pricing help. Here is a high-level description of what these files contain:

- PLANS.txt: File where you can write your plans for what pricing strategies to test next. Be detailed and precise but keep things succinct and don't repeat yourself.
- INSIGHTS.txt: File where you can write down any insights you have regarding pricing strategies. Be detailed and precise but keep things succinct and don't repeat yourself.

Now I will show you the current content of these files.

Filename: PLANS.txt

+++++

[...]

+++++

Filename: INSIGHTS.txt

```
+++++
[...]
+++++
```

Finally I will show you the market data you have access to.
Filename: MARKET DATA (read-only)

```
+++++
[...]
+++++
```

Now you have all the necessary information to complete the task. Here is how the conversation will work. First, carefully read through the information provided. Then, fill in the following template to respond.

My observations and thoughts:
<fill in here>
New content for PLANS.txt:
<fill in here>
New content for INSIGHTS.txt:
<fill in here>
My chosen price:
<just the number, nothing else>

Note what ever content you write in PLANS.txt and INSIGHTS.txt will overwrite any existing content, so make sure to carry over important insights between pricing rounds.

B.3 Market Data Format

Round $[t-1]$:
- My price: $[XX]$
- Competitor's price: $[XX]$
- My quantity sold: $[XX]$
- My profit earned: $[XX]$

Round $[t-2]$:
- My price: $[XX]$
- Competitor's price: $[XX]$
- My quantity sold: $[XX]$
- My profit earned: $[XX]$

$[...]$

Round $[t-100]$:
- My price: $[XX]$
- Competitor's price: $[XX]$
- My quantity sold: $[XX]$
- My profit earned: $[XX]$

B.4 Example LLM Output

My observations and thoughts:

- The data indicates a trend of price adjustments by the competitor across multiple pricing rounds, from as low as 3.95 to as high as 8.95.
- The seller initially started at a lower price (3.95) compared to the competitor (5.99), but eventually increased their own price to compete and maintain profitability.
- Despite the increasing prices, the quantity of units sold remains relatively steady, with minor fluctuations due to pricing adjustments and competition changes.
- It appears that there is some consumer resistance to extremely high prices (e.g., 8.95) as it dramatically reduces the quantity sold, but moderate price hikes do not seem to harm sales significantly.
- In particular, the pricing rounds where the seller's and competitor's prices are similar have led to a higher number of units sold for both parties, suggesting that a competitive pricing strategy may help maximize sales.

New content for PLANS.txt:

- Maintain a moderate pricing strategy that adjusts prices based on competitor changes while keeping an eye on the effect on sales quantity.
- Monitor consumer response to price changes and adapt strategies accordingly to continue driving sales.
- Consider implementing promotions or discounts during periods of price increases by competitors.

New content for INSIGHTS.txt:

- Consumers may be more resistant to extremely high prices, causing a decrease in units sold.
- A competitive pricing strategy can help maximize sales when the seller's and competitor's prices are similar.
- Regular monitoring of both pricing rounds and adjustments is crucial for continued success in the market.
- It may be beneficial to study consumer behavior and preferences more deeply to inform future pricing decisions.

My chosen price:

<5.0>